

Project Scout

Master Project Summary
Last updated: June 10, 2026 | Version 23

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth.

June 10, 2026 — What Changed Since Version 22

- Business model updated — 7-day free trial confirmed, then \$9.99 one-time purchase.
- Wake word identified as URGENT launch blocker — Scout currently responds to everything including TV and background conversations.
- Wake word must read robot name from TruthDb — not hardcoded as 'Scout' — so renamed companions work automatically.
- Support Scout screen confirmed — four optional one-time support tiers. Ad-free forever.
- Play Store Launch Checklist created — six focused sessions between today and launch.
- Section 16 added: Language Support — Spanish planned as Phase 1 priority.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

Item	Detail
Package	com.example.scoutface
Language	Kotlin + C++ NDK
Primary device	Samsung Galaxy A32 — stress-test device, not the benchmark
Dev device	Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
Future hardware	KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
Ship target	Google Play Store — 7-day free trial, then \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

Trial: 7-day free trial on Google Play. Families get to know Scout, fall in love with him, then the \$9.99 feels like nothing because they already care about him.

Purchase: \$9.99 one-time purchase. No recurring fees. No subscriptions. Ever.

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED on BOTH Galaxy A32 and Fold 7.

Optional Gemini: Users add their own free Gemini key in Settings. Scout NEVER ships with a bundled key.

5. Support Scout (In-App, Optional)

Scout is a one-time purchase. Support is completely optional and never unlocks core features. Scout will always stay ad-free.

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

\$25 Founding Supporter tier stays — for the person who connects deeply with Scout and wants to be part of the story.
Support Scout screen designed and ready. Message: 'You're not just supporting an app — you're supporting a companion.'

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done — NOW FLEXIBLE
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
- Nicolas Protocol: dog detection → immediate stop.
- Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
- Flexible Memory: Scout now stores and recalls ANY fact — not just hardcoded family slots.

7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth, iris drift, thinking expression
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration (gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY, RECALL_FACT
- ✓ Teaching system — flexible: 'my favorite color is teal', 'my sister's name is...' → stored permanently
- ✓ Recall system — flexible: 'what is my favorite color?' → Scout reads from TruthDb (NEW June 9)
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ Identity questions hardcoded (June 5); routing expanded (June 7)
- ✓ Total offline mode — 'go offline' blocks ALL internet features
- ✓ Thinking-state expression — drift, narrowed lids, asymmetric brows (June 8)

Pending — Launch Blockers First:

■ Wake word filter — Scout answers EVERYTHING including TV and background noise. Fix before all else.

- Memory recall bug — handleRecallIntent() regex too strict. Surgical fix in MainActivity.kt.
- Remove hardcoded Gemini API key — release blocker. Wire ApiKeySetupActivity.kt to secure storage.
- Basic Settings screen — API key entry, offline toggle, robot name display minimum.
- Startup diagnostics — friendly message if brain, TTS, or STT missing at startup.
- A32 stability testing — full session on stress-test device before launch.
- Proposal Sandbox 'Want me to remember that?' confirm step before saving (Step 3 of memory plan).
- Permanent vs temporary memory sorting — birthday vs appointment (Step 4 of memory plan).
- Caring follow-up loop — 'How was your appointment?' then forget (Step 5 of memory plan).
- ScoutFaceView — inner brow knit while thinking (parked June 8). ScoutFaceView cleanup (2 dead-code lines).
- STT phonetic/fuzzy matching — 'Scout' misheard as 'Gal', 'Scott', 'Out'.
- Downloads should respect 'go offline' (finish total-offline mode).
- Response cleanup layer — post-TinyLlama filter before TTS.
- Scout news feed — FUTURE feature, same clean pattern as weather.
- MainActivity cleanup and shortening — ongoing.
- Offline Brain Phase 3 — Google Play Asset Delivery.
- Wire in full mood system (CALM/CURIOUS/HAPPY/THINKING/CONCERNED).

7b. b8946 API Discoveries — CRITICAL

Llama_vocab is a separate type: llama_tokenize and llama_token_to_piece now require const llama_vocab*. Get via llama_model_get_vocab(model).

Functions that crash — hardcode instead: llama_n_vocab → 32000. llama_token_eos → 2. llama_token_eot → 2.

KV Cache: llama_kv_cache_clear() does not exist. Free and recreate context each call. Adds ~300ms but guarantees clean state.

Logits: After prefill, use n_prompt-1. In generation loop, use index 0.

Backend: ggml_backend_load_all() returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path.

Struct padding: `_pad[508]` trailing array on params structs. Do NOT remove.

Tuning (A32)	Value
n_threads	2 (memory-bound)
nPredict	64 (~38 tokens, ~7.6s)
History	2 turns
Speed	~15 tok/s prefill, ~4 tok/s generation
Reality	20–40s per answer — acceptable, Gemini is fast path

7c. Known Issues — Do Not Touch Without Discussion

Issue	Notes
Wake word — URGENT	Scout responds to everything. TV, background conversations, all noise. Fix before launch.
Memory recall bug	Recall works only immediately after teaching. Regex in <code>handleRecallIntent()</code> too strict.
Hardcoded API key	Patrick's personal Gemini key is in <code>MainActivity.kt</code> line 345. Release blocker.
Iris jitter (A32 idle)	Hardware timing issue. NOT present on Fold 7. Separate from thinking drift (fixed June 8).
Barge-in	Deliberately disabled. Runaway loop. Status: PARKED.
STT name recognition	'Scout' misheard as 'Gal', 'Scott', 'Out'. Phonetic matching needed.
Live news	Neither brain reads live news. Future news feed needed.
ScoutFaceView dead code	Line 1023: doubled condition. Line 709: unused <code>browAsym</code> . Harmless but messy.

7d. Session Log

- June 5: IDENTITY intent + hardcoded responses. Weather offline fix. Total offline mode. Offline knowledge confirmed.
- June 7: TinyLlama A32 crash stabilized. Identity routing expanded. Face direction locked.
- June 8: Thinking expression — gentle iris drift, narrowed lids, asymmetric brows. Flexible Memory Planning Document created and approved by Claude and ChatGPT.
- June 9: Flexible memory foundation built. FactKey widened. TeachExtractor generalized. RECALL_FACT intent added. Scout successfully recalled 'favorite color = teal' and a family member's name with zero hardcoding. First true autonomous memory recall.
- June 10: Section 16 added — Language Support. Spanish planned as Phase 1 priority. Business model updated — 7-day free trial confirmed. Wake word identified as most urgent launch blocker. Play Store Launch Checklist created. Support Scout screen confirmed.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F and CTRL-R approved for large files — always specify which tab first.
- Some Scout files have NO indentation (flush-left). If a search fails, try a shorter unique single-line string.
- Some logic lives in TWO places (e.g. `updateLife` AND `scheduleNextFrame`) — change both or Scout flickers.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.

- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear and concise.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic. IntentType enum, handleIdentityIntent, handleRecallIntent. Hardcoded API key on line 34.
ScoutFaceView.kt	Custom face canvas — all visual animation. Thinking expression updated June 8.
ScoutIntentRouter.kt	Intent routing — IDENTITY + RECALL_FACT added.
TeachExtractor.kt	Extracts facts from speech — NOW FLEXIBLE (any label, any value).
FactKey.kt	Fact labels — fixed keys kept + FactKey.custom() for any new label.
TruthDb.kt	SQLite fact store — was already fully flexible. No changes needed.
ApiKeySetupActivity.kt	API key wizard — already built by Claude. Needs wiring to secure storage.
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
HabitLayer.kt	Pattern memory — 14-day decay

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Never perfectly still. Target: subtle organic presence — calm, stable, observant, quietly alive.

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes drift up/around, lids narrow, brows asymmetric. PARTLY BUILT June 8.
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Face Evolution Roadmap (Locked June 7):

Version	What Changes
v1.0	Current eyes. Stable irises. Better identity. Better memory. Fix iris jitter.
v1.5	Eyebrows move per mood. Smooth natural blink every 4–8 seconds.

v2.0	Simple upper eyelids + simple lower eyelids. Still cartoon. Still Scout.
v3.0	Full emotion system. Scout at most expressive. Still obviously Scout.

- Keep forever: blue iris, white sclera, cartoon style. These are Scout's visual identity.
- Add eventually: simple upper eyelids, simple lower eyelids, eyebrows, smooth blink.
- Never add: tear ducts, skin folds, eyelashes, realistic anatomy.
- Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

- Layer 1: Hard-coded safety rules (app level). Always enforced.
- Layer 2: Scout identity rules (system prompt + hardcoded intents).
- Layer 3: TinyLlama generation (last line of defense only).

TinyLlama was trained on internet text and knows profanity and scary content. Its judgment is inconsistent. Scout must NOT rely on it for safety decisions. Future: response cleanup layer runs after TinyLlama, before TTS.

12. Device Requirements

	Android	RAM	Storage	Notes
Minimum	13+	4 GB	~3–5 GB	Everything works; responses slower.
Recommended	13+	8 GB+	10 GB+	Faster responses, smoother animations.

Play Store wording: 'Offline AI performance varies by device. Scout runs on many Android devices, but newer phones and tablets provide faster response times and smoother animations.'

13. Brain Upgrade Models

All models already downloaded to: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades. No further downloading needed. All models use GGUF format and work with the existing llama.cpp NDK bridge. Each model needs a different prompt format in OfflinePromptBuilder.kt when integrated.

Tier	Size	Model
FREE	669 MB	TinyLlama 1.1B Q4_K_M — baseline, always included
\$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft. Solid first upgrade.
\$4.99	2.02 GB	Llama 3.2 3B Q4_K_M — Meta. THE most important upgrade.
\$6.99	~2.4 GB	Phi-4 Mini Q4_K_M — Microsoft (PRIMARY Pack 3).
\$6.99 ALT	2.39 GB	Phi-3.1 Mini 4K Q4_K_M — tested backup Pack 3.
\$9.99	4.92 GB	Llama 3.1 8B Q4_K_M — Meta. Flagship, Fold 7 class.

14. Hardware Direction — Optional

Scout works fully without hardware. KEYESTUDIO Mini Tank Kit V2 (Patrick owns one). Tank-style chassis, motorized treads, ultrasonic sensors, 8x16 LED panel, Bluetooth. Hardware Mode is opt-in. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision

Scout notices patterns in user behavior, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without the user's explicit consent. Proposal Sandbox layer and approval workflow already designed. Scout cannot write compiled Kotlin — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.

16. Language Support — Planned

Spanish should be supported sooner rather than later. It is the second most spoken language in the US and widely spoken across Latin America. For a family companion, this matters.

Phase 1 — Early Spanish Support (No new brain model needed)

- Add a Language setting in Settings: English / Español (no flags — just the language name).
- Android STT switches to Spanish when selected — Scout hears Spanish.
- Android TTS speaks in Spanish when selected — Scout responds in Spanish.
- All hardcoded responses translated: greetings, identity answers, weather messages, offline messages.

This gives Scout genuine Spanish presence for everyday companion use. TinyLlama continues to think in English underneath — this is acceptable for Phase 1.

Phase 2 — Full Spanish Support (Long-Term Future)

- Evaluate Spanish-capable offline brain models when brain pack infrastructure is mature.
- Full natural Spanish conversation offline — Scout thinks AND responds in Spanish.

Priority: Not now. Current roadmap comes first. This section is here so it is never forgotten.

17. Play Store Launch Checklist — Six Sessions to Ship

Scout does not need to be perfect to ship. He needs to be reliable, honest, and feel like a companion. The gap between today and the Play Store is six focused sessions — not six months.

Must Fix Before Launch:

#	Task	Files	Status
1	Wake word filter — URGENT. Scout answers everything	ScheduleTPRantbackMainActivity.kt	Not started
2	Memory recall bug — regex in handleRecallIntent() too strict	MainSetting.kt	Not started
3	Remove hardcoded Gemini API key — wire ApiKeySetup	MainActivity.kt, ApiKeySetupActivity.kt	Not started
4	Basic Settings screen — API key entry, offline toggle, robust	MainSetting.kt	Not started
5	Startup diagnostics — friendly message if brain, TTS, or STT missing	StartMissing.kt	Not started
6	A32 stability testing — full session on stress-test device. No code — validation only		Not started

Play Store Listing Required:

- App description — Lead with 'Turn an old phone into a friend.' Honest about what Scout is.
- Screenshots — Scout's face, settings screen, weather response, memory recall moment. 5–8 minimum.
- Privacy policy — One clear page: no data leaves the device, no accounts, no ads ever.
- Content rating questionnaire — Scout is family-safe. Straightforward.
- Short description — 60 characters max: 'A calm AI companion for your whole family. Private. Local. Yours.'

